Palm OS Programming Recipes

Exchange Manager: Sending Records

You can send data to any PIM application as long as the type information in the `ExgSocket` structure matches the registered file type of that PIM application.

If you don't know how to use the Exchange Manager, please refer to other recipes for the Exchange Manager. The SDK documentation also provides helpful information.

The challenge of sending records--that is, structured data--using the Exchange Manager is figuring out how to send and receive structures instead of plain text. This is often an application-specific issue. The sample program that goes with this recipe uses the ToDo PIM application as an example. Therefore, you should be able to use a couple of the functions from the ToDo application.

Follow these steps:

In Constructor:

- Add a menu item for sending or for any other UI components that are applicable to your application.

Open the source file (`*.c`) and find the event loop handling function that will handle the "Send" or "Beam" menu item.

- Make sure that `ExgMgr.h` is included as a header file.
- Create and initialize an `ExgSocketType` data structure with information about which library to use and the data to be sent. Make sure to initialize the description, type, target, and/or name.
- Set `name` to end with .vcs. ToDo is registered to receive this file type.
- Call the `ExgPut` function to establish the connection with the exchange library.
- Send the data out. This task is application-specific.
- Call the `ExgDisconnect` function to end the connection.

These code segments are examples of the above procedure:

```
static void SendToLocalMemopad()
{
     ExgSocketType exgSocket;
     Err err = 0;
     UInt32 size = 0;
     UInt32 sizeSent = 0;
     Int16  recordNum = 0;   // send the first record in the ToDo database
     DmOpenRef dbP = NULL;
     ToDoDBRecordPtr recordP;
     MemHandle recordH;

     // Find the database for ToDo application
     dbP = DmOpenDatabaseByTypeCreator ('DATA', sysFileCToDo,
        dmModeReadWrite);
```

```
        // won't happen, put this here just in case
    if ( dbP == NULL ) return;
        // get the first record of the ToDo application
        recordH = (MemHandle) DmQueryRecord(dbP, 0);
        // if there are no record in ToDo database, exit
    if ( recordH == NULL )
    {
        // alert user for empty todo list database
        FrmAlert (ToDoListEmptyAlert);
        // but remember to close the database
        DmCloseDatabase(dbP);
        return;
    }
    recordP = (ToDoDBRecordPtr) MemHandleLock(recordH);
    size = MemHandleSize(recordH) + 100;

    // important to init structure to zeros
    MemSet(&exgSocket,sizeof(exgSocket),0);
    exgSocket.description = "ToDo item from ExgMgrSendRecrod";
    exgSocket.type = "text/x-vCalendar";
    // set target, todo and date book share an extension
    exgSocket.target = sysFileCToDo;
    // do this instead if you use 4.0
    // exgSocket.name = "?_beam:SendRecord.vcs";
    exgSocket.name = "SendRecord.vcs";
    exgSocket.length = size;

    err = ExgPut(&exgSocket);
    if (!err) {
        // the way to handle ToDo database record in 3.x
        // note that 4.0 had a complete new way to do this by using
        // PDI lib APIs
        PutString(&exgSocket, "BEGIN:VCALENDAR"
              imcLineSeparatorString);
        PutString(&exgSocket, "VERSION:1.0" imcLineSeparatorString);
        ToDoExportVCal(dbP, recordNum, recordP, &exgSocket,
              PutString, true);
        PutString(&exgSocket, "END:VCALENDAR" imcLineSeparatorString);
        MemHandleUnlock(recordH); // free the memory handle
        DmCloseDatabase(dbP);   // close db every time in the eventloop
        ExgDisconnect(&exgSocket,err);
    }
}
```

Sample program downloads:

- Windows
- Mac

As a demonstration, the sample code that comes with this recipe implements shortcut-B for beaming and command bar for beaming, which are standard for beaming. Please check the sample code for details.